

Towards Enhancing Empathy Through Emotion Augmented Remote Communication

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ABSTRACT

Our ability to empathize stands at the core of our existence as humans. It has dramatically contributed to our evolution as species and remains key to our relationships with family, friends, and colleagues. Yet, we sometimes find it difficult to empathize, more so in the online space increasingly dominating our interactions.

In this work, we explore the potential of videoconferencing platforms as avenues for augmenting empathy. We have developed Project Us, a system that uses machine learning to analyze interlocutors' tone of voice and facial expressions during online interactions, compiles their emotional valence and feeds it back to each participant as a real-time visual cue. Tested with 40 users, Project Us shows the emergence of empathy-related behaviours, such as a higher awareness of own and other's emotions, and an increased use of pro-social language, as evaluated through quantitative (natural language processing) and qualitative methods. We also analyze ethical challenges of empathy augmentation.

CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI)**; *Collaborative interaction*; *User studies*.

KEYWORDS

Augmented Communication; Empathy; Facial Expressions; Speech

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Figure 1: User Interface of Project Us. A tone of voice or a facial expression with a strong negative valence triggers a temporary change to color orange in the corresponding circle.

1 INTRODUCTION

There is growing evidence showing that empathy plays a significant role in the quality of our relationships. It can make us better family members [19], friends [28], managers [27] and colleagues [15]. In turn, this improved relationship quality can be a strong driver of our self-esteem and well-being.

Yet, this experience is often incomplete. We often fail to empathize with people around us, and for poorly understood reasons, some people are naturally more empathetic, while others face significant challenges [32].

The idea that empathy can be learned is not completely new. Over the past decades, empathy training has started being used as therapy, particularly for children with ASD (Autism Spectrum Disorder) [11]. At the same time, the recognition of its value in jobs that heavily rely on human interaction, from education and healthcare to management and customer service, has earned empathy a place in many training curricula.

Technological solutions have also started to emerge (e.g., perspective taking based on Virtual Reality [17] or films augmented with physio-signals [33]); however, many existing solutions still rely on specialized lab settings or dedicated equipment, posing limitations in their scalability and accessibility to a broader population. [6].

With Project Us, we aim to provide an accessible, easy-to-use tool that helps people augment their empathy. The tool analyzes a set of core signals of people engaging in an online interaction

(tone of voice and facial expressions) with the help of two machine learning algorithms, in order to extract information about their emotional valence. It then creates a continuous feedback loop (in the form of a subtle on-screen display), making each user aware of both their own and their partner's evolving emotions. We define empathy as cognitive empathy (sometimes known as 'perspective taking'), meaning that one looks at the world from the other's point of view.

Even though the algorithms can obtain more granular information (e.g., indications of "angry" and "happy"), Project Us only displays aggregated cues about emotional valence. This is because it does not aim to read emotions, but rather to initiate a learning process by stimulating users to corroborate the feedback with their own observations. The tool is compatible with any commercial communication platform (e.g., Zoom, Skype, Google Hangouts, etc.) without the need for specialized hardware or a physical setup.

In this paper, we describe the main related work (Section "Related Work"), our design choices (Section "Design"), and the methodology and results of a user study testing Project Us' impact and functionality when relying on speech and tone of voice (N=20) and facial expressions (N=20) as input signals (Sections "Methodology" and "Results").

The studies relied on a survey that the participants were asked to respond to at the end of each experiment, as well as the post-interaction analysis of the conversations carried over with and without the activation of Project Us. The user feedback indicated increased levels of reflection (defined as the tendency to analyze others' emotional experience) and empathy-related actions (i.e., behaviors aimed at easing the interaction, such as asking questions or changing the tone of voice). Users' comments also suggested curiosity towards using the tool in real life, many finding it easy and comfortable to use. At the same time, multiple users surfaced the need to pay special attention to issues of privacy and consent, and some provided insightful design suggestions (e.g., the use of a more fluid visualization instead of circle, given the fluid nature of emotions). The post-interaction analysis (using Natural Language Processing - NLP, specifically Linguistic Inquiry and Word Count analysis) also indicated changes in the choice of language (e.g., towards collective nouns, empathy-related words).

In the Discussion section, we will compare the results of using different input signals, as well as elaborate on challenges related to their corroboration. In the section Ethical Considerations we will discuss potential issues related to the misuse of the tool, privacy, technological authority, and our mitigation strategies. As described in the Conclusions and Future work section, our next phase of the project will focus on exploring the combination of input signals, further design of the feedback visualization, as well as the longer-term evolution of the observed behaviors.

2 RELATED WORK

Enhancing our ability to empathize has been the subject of research for multiple decades.

Traditional approaches include didactic training (e.g., a facilitator-led class focused on concepts and best practices) [33] [22] and online tutorials (e.g., multimedia resources illustrating hypothetical situations and advice on how to handle them) [37] [21]. The effectiveness

of such approaches has nonetheless proven limited, especially due to their disconnect from learner's real life context. Personalised coaching has in turn shown higher effectiveness, yet a more limited accessibility to the broader population.

Virtual reality (VR) has recently emerged as a promising technologically-enhanced solution. Using the illusion of embodiment, it allows users to "walk in someone's shoes", thus encouraging empathetic behaviors [9] [25] [44] [4] [40]. A theoretical analysis of mechanisms of empathetic behavior associated with VR can be found in [4]. Nonetheless, unlike Project Us, VR solutions continue to rely on hypothetical situations, a potential limitation to their effectiveness.

Other approaches rely on wearable devices. For example, Google Glass can support the therapy of children with ASD (Autism Spectrum Disorder) [11] [43] [10]. Another example is [34]- a set of wearable bracelets that measure users' EDA (electrodermal activity) and deliver real-time emotions-related feedback during face-to-face conversations. While not focused on empathy, Neo-Noumena [36] and [20] use EEG and Augmented reality headsets to share affective information and support emotional co-regulation and attention (respectively). By only relying on widely available hardware (e.g., a laptop), we hope Project Us can become accessible and easily scalable to a broader population.

Another set of related solutions (often enabled by wearables) aim to augment communication by sharing, in real time, raw physiological data (e.g., heart rate, respiration rate, EDA). They have also shown to have an impact on empathy-related behaviors [18] [23] [9] [46] [45] [14] [16] [12] (an extensive review can be found in [26]). With Project Us, we do not aim to share raw biosignal data, but insights on its emotional content. Our choice could potentiate the empathy enhancing effect, but also raises new design, usability and ethical challenges, given the more prominent role of a computer as a mediating agent.

Potentially closer to our work, other research explores the use of emotions feedback in remote settings. [39] shows mood feedback can aid remote collaboration. This feedback nonetheless relies on the observations of a human mediator. [7] also uses emotions-related feedback based on audio analysis to help customer representatives enhance their communication and empathy. In addition to the analysis of facial expressions, Project Us also differentiates itself through its aim to build a "humans in the loop experience", where both participants benefit from feedback on their and their peer's emotional state. Other approaches aim at modifying communicative elements (e.g., facial expressions, voice, etc.) to study the effects on communication. For example [31] [24] use real-time facial/voice deformation or avatars in videocalls to enhance creativity [30] or engagement [41]. Our work is nonetheless exploring a different learning mechanism, based on subtle nudges that can potentially lead to more sustainable behavioral change [5].

3 RESEARCH QUESTIONS AND HYPOTHESES

Through designing, developing and testing Project Us, we have aimed to answer the following research question: How can a real-time feedback loop about the emotions of participants during an online conversation, as analyzed from their tone of voice and facial expressions, affect their empathetic behaviors?

To study this question, we carried out a within-subject experimental study to compare the emergence of empathetic behaviors (e.g., awareness of peer's emotions, actions to improve the interaction) in conversations with and without feedback (both participants see the same feedback, i.e., on themselves and on the peer). A between-subject design is more sensitive to variability between subjects, hence not considered at this stage. Our hypothesis is that the conversations enhanced with real-time feedback display more frequent empathetic actions, than the ones without.

4 DESIGN

Project Us was designed as a software that can run in parallel to any video conferencing system (e.g., Zoom, Microsoft Teams, Google Hangouts), without requiring specific add-ons. The software 1) Captures each individual's tone of voice and facial expressions 2) Processes them on with two machine learning algorithms to extract cues about users' emotions and 3) Shares real-time feedback in the form of a visual cue with participants when it detects a negative emotional valence. Both participants receive feedback about themselves and their peer.

The tone of voice and facial expressions from each user are fed in real time into two machine learning models that extract information about the conveyed emotions by analyzing the acoustic characteristics of the voice (Open Vokaturi ML algorithm, accuracy 76.1%, computes the emotion probabilities via a neural network with three levels of linear connections [35]) and of the facial expressions (Face-API ML algorithm, accuracy 70-80% [38], obtains the emotion probabilities via depthwise separable convolutions and densely connected blocks [29]). Combining the two types of input is out of the scope of this paper (see Section 7 for details). The algorithms classify the emotional cues into one of the following categories:

- Tone of voice: anger, sadness, happiness, fear, neutrality
- Facial expression: anger, sadness, happiness, fear, disgust, surprise and neutrality

FaceAPI and Vokaturi output each a vector with the classification probabilities for each emotion (period 1s). In case the system detects a negative valence in the emotional experience of one user (i.e., predominating anger, sadness, fear or disgust; with a probability above 0.5/1), it feeds it back to both of them in the form of a visual cue (orange circle) on the visual interface. If the valence is neutral or positive, the circle remains white (see Figure 1). We have chosen the orange color instead of red after multiple early users shared their perception of red as "aggressive" and "distracting".

The participant views two circles (Figure 1), which display emotion cues from themselves (left) and from the peer (right). Therefore, both participants have access to the same information. We include both circles to study the effect of emulating a humans in the loop context, by dynamically (and as symmetrically as possible) combining two core empathy enhancing triggers – self-awareness (own circle) and awareness of others (peer's circle).

We chose the circle format with a color transition to design the cue as simply and intuitive as possible. We converged on this type of feedback upon conducting early experiments with two alternative models, namely:

- Real-time emotion-specific colored circles: each type of emotion detected was represented by a different color of the on-screen circle (for both self and peer)
- Cadence of own and peer's emotion: the interface was showing a timeline with the evolution of each peer's emotional valence throughout the call

Both options were deemed distracting by early users, given the cognitive load of analyzing multiple data points. Users described a temptation to enter a "guessing game". While we acknowledge this could in itself result in a learning effect, for the purposes of the initial design, we have chosen to start by analyzing the impact of a simple cue in order to minimize distractions from the conversation. Nonetheless, we decided to display both circles (about oneself and about the peer) to both participants, in order to emulate the complementary self and peer awareness.

5 METHODOLOGY

In our implementation of the latest design, we tested Project Us with a total of 40 users (20 pairs), split into two cohorts: 20 users (10 pairs) whose tone of voice was analysed and 20 users (10 pairs) for whom we analyzed the facial expressions. There were two reasons for splitting participants into cohorts. On the one hand, we wanted to explore potential differences in the use of the two inputs and their perceived impact on conversation. At the same time, we were still working on devising a solution for fusing the emotional information extracted from the two inputs (tone and facial expression) in a way that they can provide complementary, or even synergistic information (e.g., how to interpret a situation where the facial expression indicates surprise, but the tone indicates anger).

Each of the 20 experiments (10 per pair) consisted of two 10 minute emotionally intense conversations between strangers. Participants visualized the experiment screen in both conversations (i.e., the entire experiment), but the emotion feedback was only displayed in one of the conversations. The conversation with no feedback triggered is the control condition. The feedback was given during the first conversation to half of the pairs, and during the second one to the other half (i.e., counterbalanced condition). We do not include a condition with random feedback in this study because we would like to validate first if the feedback has an effect. As a next step, the random feedback could help us further understand the triggers of observed behaviors (incl. potential placebo effects).

The conversation topics sought to elicit intense emotional reactions in the participants, even though they were strangers. The topic was selected by experimenters upon asking each participant to rank a list of ten "controversial topics in the U.S." (e.g., "Should people who disregard COVID rules be fined?" or "Should affirmative action be abolished?") in decreasing order of importance. Then, participants ranked the topics (agree/disagree) on a scale from 1-10. The experimenters then paired participants on topics that they had ranked highly, but on opposite ends of the spectrum.

We have chosen to share with participants the type of input that was being analysed: tone of voice or facial expression, given some participants' explicit interest, as well as to avoid the risk of raising suspicions. We have also explained that the system has limitations and some of the feedback might not be correct. The purpose of this

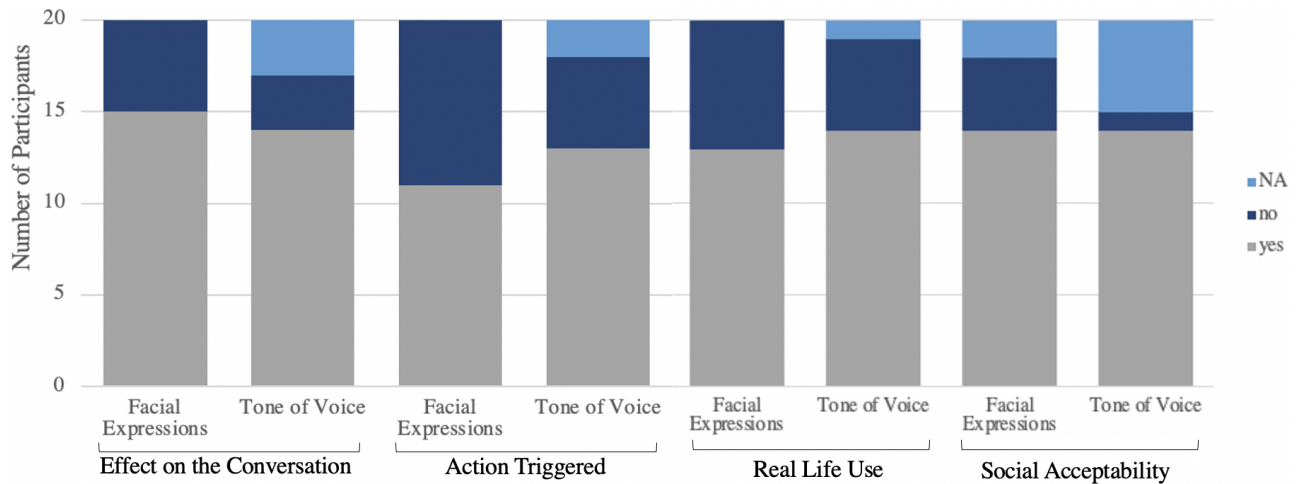


Figure 2: (Left) Reports from participants about the tool’s effect on the conversation. In all cases, most participants declared the feedback affected the conversation and that they took an action in response (comments in Section 6.1.1). (Right) Reports from participants about their intent to use the tool and its social acceptability. In all cases, most participants declared they would use the system in real life and deemed it socially acceptable (comments in Section 6.1.2.)

was to promote an effort from the users to corroborate the feedback from Project Us with their own observations. We expect this to support learning and discourage users from “blindly” relying on the computer’s feedback.

The experiment included 40 participants (20 pairs, 20 males, 20 females). Conversations were held in English. Each of the interactions was audio recorded and transcribed. At the end of each series of two conversations, participants completed a survey with open and closed questions aimed at analyzing the effect of the feedback, as well as the way users experienced the tool, and their interest in continuing to engage with it. Upon completion of the experiments, the facial expressions and tone of voice recordings (with and without Project Us feedback) were analyzed based on LIWC: Linguistic Inquiry and Word Count analysis to explore potential changes in the choice of language.

User studies were done after approval from the Review Board of our institution.

6 RESULTS

In this section, we will share the results obtained from the user survey performed at the end of each conversation, as well as a series of post analyses of the data obtained from the conversations.

6.1 User surveys

At the end of each experiment, we engaged each participant in a short survey. This served three purposes: i) explore any empathy-related behaviors observed by users themselves, ii) collect feedback on the user experience with Project Us (e.g., level of comfort) and intent of use, and iii) surface potential concerns related to the system and its social acceptability.

6.1.1 Empathy-related behaviors observed by users. As shown in Figure 2, most participants felt the feedback had an impact on the conversation, slightly more in the case of the facial expression input, as compared to the tone of voice (15/20 and 14/20 respectively).

Beyond this, most users also declared having taken an action following the feedback (12/20 in the case of the facial expression and 13/20 when their tone was being analysed).

The types of actions taken can be structured into two categories: Reflect and Act. Using the definitions from [34], Reflect refers to “*the analysis of the emotional experience of the interlocutor*” (e.g., thinking about own comments, evaluating their impact on the partner, etc.). Act refers to “*triggering a behavior aimed at easing the interaction*”.

Reflect Several participants reported that the system increased their level of awareness of the emotions they were conveying and of their effect on the partner. During the conversations with tone of voice as input, Diana (pseudonym) stated “*I wanted to be cautious of how my voice conveyed emotion. I didn’t want my partner to think I was upset or angry because I really wasn’t.*”. Joe shared that “*It helps to see what others are feeling. I am not the greatest at recognizing that myself. It made me more aware that I should be paying attention to others’ emotions.*”. Henry explained that he used the system to identify “*if my partner was actually disagreeing or agreeing with me*”. While the facial expressions were being analyzed, Matei (pseudonym) shared that the feedback “*Made me reflect on whether I had said something that might have upset my partner and think more before my next response*”. Corina shared that “*The feedback actually confirmed my assumptions about how he felt*”. Ash also added that “*The feedback made me ask myself how she was perceiving me, but also made me stop to watch her*”.

Act Most participants also reported taking some form of action when they noticed the visual feedback. During the conversations with tone of voice as an input, Atsuko explained that “*I attempted to soften my voice, speak slower, and choose my words more carefully*”, while Javier shared that “*I paused and tried to speak at a lower, more neutral register while maintaining a smile to show that I was feeling happy to be in the conversation*”. Additionally, several participants declared that they tried to use the system as an aid for the conversation. Joanne shared that “*When the circle turned red for my partner, I tried to come up with a better argument*”. While facial expressions were being analyzed, Mark mentioned that “*I tried to*

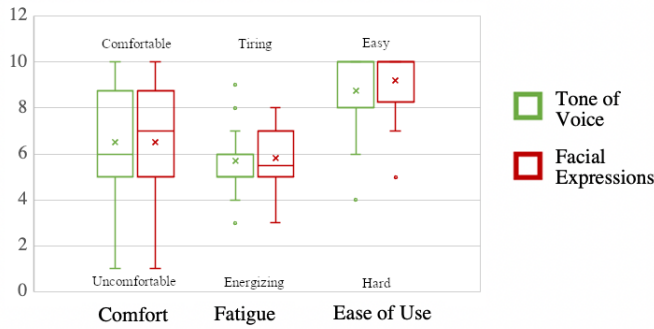


Figure 3: Box-plot with the usability results using a 10-point Likert scale to rate Comfort (uncomfortable - comfortable), Fatigue (Tiring - Energizing) and Ease of use (Easy - Difficult). 20 participants used the Tone of Voice input and 20 used the Facial Expressions input.

look more positive", but Camila mentioned that she actually "Tried to neutralize my facial expression". Virginie added that "I tried to slightly alter both my face and tone".

6.1.2 User experience. In addition to studying the impact of Project Us on the actual conversation, we also gauged users' experience while engaging with Project Us, and views on whether and when they would consider using it in real life.

Most participants declared they found Project Us easy to use, comfortable, and with no impact on fatigue (see usability results in Fig. 3), we have not observed differences in this result with the two inputs studied (i.e., we have not found statistically significant differences between the conditions "tone of voice" and "facial expressions", one-sided paired two-sample t-test, $p > 0.05$). Nonetheless, even in this significantly simplified choice of feedback, some participants did mention that the feedback was occasionally distracting them from the conversation. At the same time, others mentioned it was easy to just have it run in the background and check it from time to time. Two comments in particular prompted us to reflect upon alternative design choices. Nina mentioned she "Found it at times difficult to monitor both circles at the same time". In future experiments we can allow participants to choose in between having both the feedback about themselves and the peer, or only the one of themselves. At the same time, Maria suggested to "Not make the feedback a circle. Emotions come and go like water or air so I think a more flowy visualization would work better". We found this an interesting area of further experimentation.

At the same time, most participants surveyed declared they would be interested to use Project Us in real life. As shown in Figure 2, the facial expressions vs. tone of voice input did not seem to make any difference (13/20, and 14/20 intent of use respectively). When it comes to instances of when they would use it, Matei mentioned that "This system would be very useful in classroom situations, especially with younger children. I can see it fostering awareness of the usage of respectful language". Mihai added "I would be interested in using it with my sister who suffers from autism and struggles with this". Toma also mentioned he "would be interested in using it when going on a date, also an online date". Other participants mentioned they could use it during job interviews, to improve their speech and

ask more questions, or when engaging with clients at work. Participants who mentioned they would not use it in real life motivated it by the fact that they already feel sufficiently able to read their peer's emotions. One participant on the other hand mentioned she would actually want more detailed feedback to find it truly useful.

6.1.3 Potential concerns. Both the tool relying on facial expressions and the one using the tone of voice were found socially acceptable by 14 out of each of the 20 users (a total of 28/40, Fig. 2). Among social acceptability concerns, participants surfaced privacy and consent. Ina nonetheless shared that Project Us "doesn't impose too much on privacy in my opinion" when discussing social acceptability, while Jack added "could be acceptable given data is used correctly". Lara highlighted the importance of consent "I believe it's socially acceptable with consent, which was the case today". This feedback will be considered as we refine our privacy practices, in light of expanding the study to larger cohorts.

6.2 Post-interaction Analysis

To further explore the impact of the Project Us feedback on the conversations, we have also performed a series of analyses of the conversations with or without feedback. While our sample size (40 participants) is relatively limited, we have aimed at identifying early patterns to inspire our future design choices. As mentioned in the previous section, half of the participants received feedback during the first conversation, while the other half received it during the second (to counterbalance the effect of the sequencing, such as participants becoming more comfortable to one another).

We have used Linguistic Inquiry and Word Count analysis (LIWC [42]) to measure indicators of attempts to understand the other, or expressions of emotion, such as "perception" (e.g., feel, see, hear). In conversations with feedback, the share of words showing perceptual processes increased, for both the tone of voice input (by 20.5%, from 16.79 to 20.24, and the facial expressions input (by 2.6%, from 16.64 to 17.08) (Fig. 4). One-sided paired two-sample t-test revealed $p > 0.05$ for both cases, i.e. not statistically significant, though we will complement it with a larger sample in future experiments.

Additionally, we have analyzed the impact of Project Us on awareness by looking into pronouns. The occurrences of "We" instead of "I" significantly increased in conversations with feedback for both the tone of voice input (from 3.5 to 4.8 occurrences per 1000 words, one-sided paired two-sample t-test $t = 3.497$ and $p = .032$), and the facial expressions input (from 2.99 to 5.97 occurrences per 1000 words, $t = 2.533$ and $p = .016$) (Fig. 4). Such changes in the use of inclusive pronouns have been used in previous studies as an empathy marker [1].

7 DISCUSSION

This section discusses the experimental results, as well as potential challenges to be addressed as we advance.

The experimental results indicate that providing feedback through Project Us during online conversations is associated with the emergence of a series of empathy-related behaviors. This can be observed consistently in all the quantitative metrics tracked using NLP, where the feedback is associated to a higher proportion of words associated with perceptual processes and an increase in the instances of the pronoun "we". On top of the changes in language

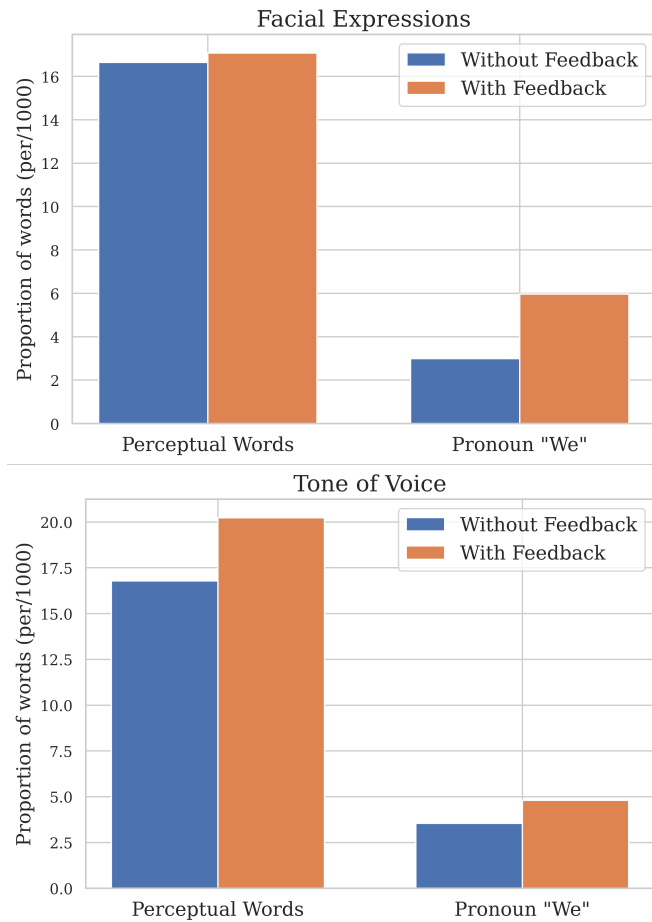


Figure 4: When using facial expressions (top) or tone of voice (bottom) as input for emotion detection, the feedback (FB) from Project Us is associated with an increase in the use of the pronoun "We" and Perceptual Words (measured in proportion of words per 1000).

analyzed, users themselves have noticed the use of Project Us to lead to some changes in their attitudes towards the conversation (Section 6.1). For example, being more aware of the peer's and their own behaviors, and taking a set of actions aimed at easing the conversation (e.g., pausing and rephrasing their points).

Tone of voice vs. facial expressions As mentioned in the methodology, part of the reason we have chosen to study the effect of the two types of input in different cohorts was the desire to study differences in the impact and user experience. The experiments nonetheless suggested very similar impacts in between the two inputs in terms of choice of language and the types of changes observed by the users themselves. Users also did not report differences in the way they experienced Project Us when their tone of voice was being analysed, versus their facial expressions.

This poses interesting questions about the effects of the feedback on the conversation, the level of accuracy needed in emotion detection to actually drive behavioral change and the actual value of combining the two inputs.

Combining Tone of Voice and Facial Expressions In the next experiments, we aim to also study the potentially synergistic

effect of combining the feedback from the two different type of inputs. Nonetheless, this poses a challenging arbitration question. In cases where the emotion detected from analyzing the two inputs coincides, there is a question of the incremental value of superposing them. But if they do not, which of them is right? How do we interpret a situation where the facial expression indicates surprise, but the tone indicates anger? While our brains are used to picking up subtle cues to analyze combinations, the automatic data fusion in emotion detection algorithms is still an open problem.

Type of feedback How to feed back the information to the users has been an important design question. As mentioned in the design section, before deciding on the type of feedback used, we also ran a set of preliminary tests with different visualizations, namely i) real-time emotion-specific colored circles and ii) a timeline with the cadence of own and peer's emotion. Both displays were deemed distracting by early test users, given the cognitive load of analyzing multiple data points. Users also described the temptation to enter a "guessing game" with the machine. We have chosen to start by analyzing the impact of a simple cue, to minimize distractions from the conversation. Nonetheless, we have decided to display both circles (about oneself and the peer) to both participants, to emulate the complementary nature of self and peer awareness / a "humans in the loop" experience.

At the same time, we acknowledge the current design of the system has a negative bias (we only provide neutral and negative feedback). In the next experiments we will thus aim to also add an explicit cue linked to the detection of positive emotions, as "no news is not necessarily good news". This observation is also consistent with the recent feedback received by Amazon's Halo system [13].

8 ETHICAL CONSIDERATIONS

Exploring the intersection of machine learning and human emotions comes with a significant ethical responsibility. In this section, we aim to highlight limitations of the emotion recognition algorithm, as well as potential misuses and data privacy concerns related to Project Us. We will also describe the mitigating actions that our team has taken and continues to develop to overcome them.

8.1 Limitations and Challenges

Inaccurate Automatic Emotion Detection. While rapidly advancing, existing machine-learning algorithms continue to display inaccuracies when classifying human emotions [8] (e.g., we used a frame-by-frame facial expression algorithm that is designed for posed faces, this can display limited accuracy in free conversation setups [2]). People show multiplicity and ambiguity of expressions, and we often require extensive context and the possibility to ask questions in order to truly understand the emotions a person is experiencing. Additionally, how people communicate emotions varies across cultures, situations, and individuals. Therefore, the signals we use for detecting emotions do not always capture sufficient information to correctly classify the emotion [3].

If unattended, these limitations could lead to undesired technological authority effects, where participants assume that the output of the emotion recognition algorithm is the ground truth, which can negatively impact their decisions and feelings. In addition, incorrect classifications have the potential to mislead both partners.

Malicious and Forced Use of Emotion Recognition. While being able to recognize other's emotions can help people become more empathetic, it can also turn into a tool of malicious use. For example, in a conversation with Eric, Ruth might test multiple delivery styles to learn how to place Eric in an emotionally vulnerable position. This might later be used by Ruth to gain leverage in work-related negotiations. Related to this, an employer may attempt to use the system for the purposes of evaluating their employees behaviors in a potentially unfair manner.

Privacy Protection. By the nature of its application, Project Us has the potential to reveal information that users might deem personal and prefer to keep private, such as their ability to respond with empathy to a difficult situation or the relations between the way they feel and their facial expressions. For example, Daniel might frown to express disapproval, but also tend to raise his eyebrows when feeling angry.

8.2 Actions

Throughout the development of Project Us, we have tried to implement a series of mitigating actions to reduce the risks described above, in addition to standard practices (e.g., seeking more diverse and inclusive datasets). We describe these actions below. At the same time, we acknowledge this is a serious set of challenges and we continue to strive to devise more effective strategies.

- (1) Co-design of the system, from the first day, with end-users and multidisciplinary perspectives - In designing Project Us, we have followed an interactive process of constantly engaging with users to understand their potential concerns and drive the design in a way that limits potential misuse. We also seek constant feedback from specialists in psychology, ethics of technology, neuroscience, education and mental health.
- (2) Limit the use of the tool to (self-learning) and training setups. Project Us is designed as a voluntary, self-growth tool. All the information captured and analyzed by the tool is only available to the end user and (with consent) to its peer, for the purpose of learning.
- (3) Include a didactic step before participants proceed to using Project Us. In this step (which takes the form of a video introduction), participants learn about the limitations of the ML tool (e.g., possible misclassifications and biases), the privacy protection policy, the procedure to interrupt participation in the experiment, and other necessary information. Only then are they asked for an informed consent about using the tool.
- (4) Actively encourage participants to be more aware of self and others' emotions and critically corroborate the tool's feedback with their own assessments. This seeks to prevent technological authority effects (assuming that the output of the ML algorithms about emotions represents the ground truth).
- (5) Including a "cooperation" step at the end of the experiment that allows participants to give feedback to one another. In this phase, participants can use the ML tool to access the emotional analysis of their conversation and can talk about and illustrate their feedback with specific examples.

The set of actions above can only be a first step towards tackling the challenges. These measures will evolve as we continue to develop, evaluate and learn about Project Us. We hope our findings will ensure the tool accomplishes its intended purpose of enhancing humans' sense of empathy, while creating a safe and pleasant experience and helping inform other research and development efforts in the field.

9 CONCLUSION AND FUTURE WORK

Empathy is a cornerstone of the human experience, yet one of the hardest muscles to train, potentially more so in an age of growing remote interactions. We have proposed, designed, prototyped and tested Project Us, a system that analyzes interlocutors' core signals during online interactions and feeds them back to each of them in the form of a discrete on-screen cue. Through a study of 40 participants, we have provided early evidence that the use of Project Us can lead to the emergence of empathy related behaviors, such as increased levels of awareness of peer and the self and actions aimed at improving the conversational experience. The aggregate study on the interactions has also indicated a pattern of changes in the choice of language. Also, user feedback has indicated a positive overall experience, as well as the intent to use the tool in real life. It has also generated insights that will be core to our next design and testing choices. Particularly, in the next iteration of Project Us, we aim to explore the following directions:

- **Training Curriculum** - Studying the effects of Project Us in an empathy training program, as well as integrating the "Didactic" and "Cooperation" steps described in Section 8.2.
- **Input signals** - Exploring the value add of combining the two types of input (tone of voice and facial expressions), and potentially others (e.g., heart rate), as well as the arbitration method in case of dissonant detection
- **Feedback form** - Exploring additional ways of providing the cue, both in terms of the information shared (e.g., addition of explicitly positive feedback) and in terms of visualization (seeking to minimize distraction and, as per the suggestion of one of the participants, exploring the display of emotions as visual flows instead of circles).

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